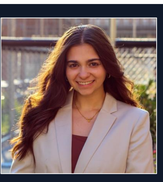
Assessing Irrigation Distribution Uniformity with UAVs in California Almonds

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Poster
References
& Contact Info

Introduction

- California is home to 80% of the world's almond production [1]
- Almonds require 101-156 cm/yr of water, but receives 25-75 cm/yr of precipitation
- Distribution Uniformity (DU) – the evenness of water distribution across a field; time-taxing assessment; poor DU wastes water
- Latent heat flux (LE [W/m2]) is the energy exchange between land and atmosphere as water changes phases; provides leaf evaporative cooling; increases as air temperature increases [2]
- Under-irrigation (and other factors) cause plant water stress, LE decreases and land surface temperature (LST) increases
- Drones can map LE, LST, and Normalized Difference Vegetation Index (NDVI)
- **So, if they're related, can we map DU using drones, too? What are the best remotely-sensed indicators of irrigation uniformity?**

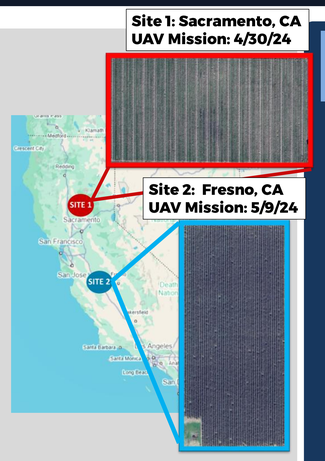




Figure 1. Poor Irrigation DU Illustration: low irrigation DU can easily cause nonuniformity in yield and plant water stress

$$DU = \frac{Average\ Flow\ Lowest\ 25\%\ Emitters}{Average\ Flow\ All\ Emitters}$$

Objective Statement

The goal is to create an easy-to-use UAV-based model to remotely test DU at two CA Central Valley orchards so farmers can test their irrigation system performance more often, avoiding unknown water waste. This project is part of the larger Tree crop Remote sensing of Evapotranspiration eXperiment (T-REX): optimizing irrigation water usage in orchards across California [1], [4].

Approach

CONCURRENT

Drone Missions
Blue, Green, Red, Red Edge, Near Infrared, and Thermal [K] bands

Meteorology & Canopy Data
Wind Speed
Air Temperature
Leaf Area Index (LAI)
Shortwave Radiation
Vapor Pressure

Irrigation Distribution Uniformity Assessment:
Measure volume in array of catch-cans

Two-Source Energy Balance Model (TSEB)
ET = latent heat flux

$$(Rn_s + Rn_c) - G = (H_s + H_c) + (LE_s + LE_c)$$

Rn = Net Solar Radiation [W/m²]
G = Soil Heat Flux [W/m²]
H = Sensible Heat Flux [W/m²]
LE = Latent Heat Flux [W/m²]
s = soil & c = canopy

Figure 2. The Two-Source Energy Balance Equation; partitions soil and canopy contributions to Rn/G/H/LE using radiometric surface temps and orchard conditions [1], [2], [3], [4]

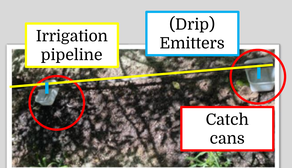


Figure 3. Irrigation DU Catch Can Test Setup: an array of catch cans set up below emitters; DU rating based on distribution of volumes captured [5]

Evapotranspiration (ET) Maps
Factors that affect ET rates include humidity, air temperature, wind speed, and plant health; site-specific analysis

VS.

Irrigation Distribution Uniformity (DU) Results
Indicate whether physical irrigation system is supplying water evenly

Irrigation Distribution Uniformity (DU) Test Results

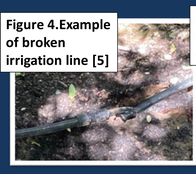


Figure 4.Example of broken irrigation line [5]



Figure 5. Functioning micro- irrigation emitter [6]

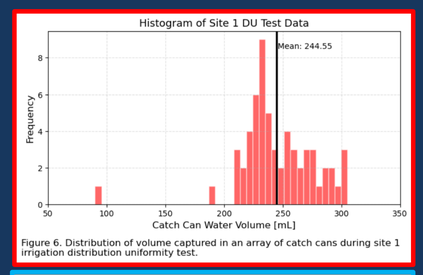


Figure 6. Distribution of volume captured in an array of catch cans during site 1 irrigation distribution uniformity test.

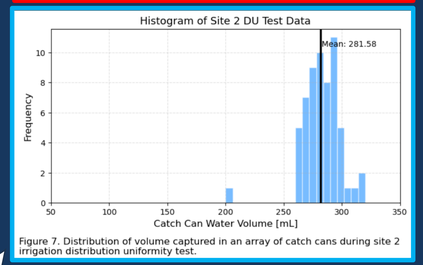
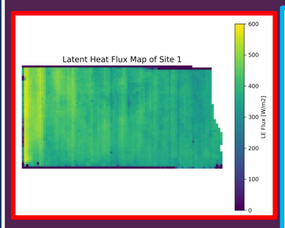


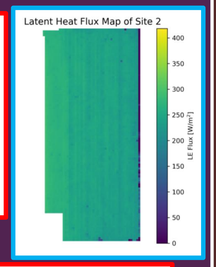
Figure 7. Distribution of volume captured in an array of catch cans during site 2 irrigation distribution uniformity test.

Table 1. Site 1&2 DU Test Result Summary		
Catch Can Volume [mL]	SITE 1	SITE 2
Avg. of Lowest 25%	208.40	263.00
All Emitters	244.55	281.58
Irrigation DU	**0.852**	**0.934**

Figures 6-7 and Table 1 show that site 1 catch can volume distribution is more spread than site 2, indicating less evenness of water distribution in site 1 compared to site 2. Ideally, the histogram would be very narrow for near perfect uniformity (e.g., site 2).

Latent Heat Flux (LE)





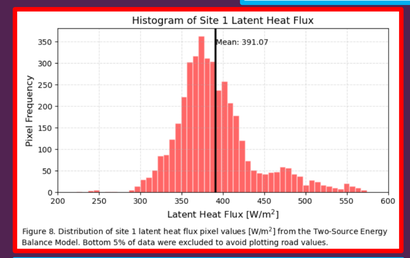


Figure 8. Distribution of site 1 latent heat flux pixel values [W/m²] from the Two-Source Energy Balance Model. Bottom 5% of data were excluded to avoid plotting road values.

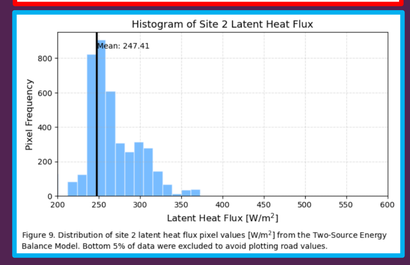
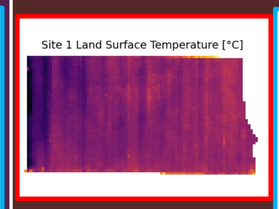


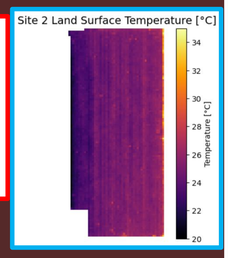
Figure 9. Distribution of site 2 latent heat flux pixel values [W/m²] from the Two-Source Energy Balance Model. Bottom 5% of data were excluded to avoid plotting road values.

Table 2. Site 1&2 LE Summary Statistics		
Latent Heat Flux [W/m²]	SITE 1	SITE 2
Minimum	222.26	315.83
Maximum	574.23	418.44
Mean	391.07	355.53
Standard Deviation	48.81	20.62

Figures 8-9 and Table 2 display the greater heterogeneity in latent heat flux values in site 1 (top) than site 2 (bottom), indicating that tree heat stress is uneven throughout the orchard. This could be due to many factors, one of which being uneven irrigation water distribution.

Land Surface Temperature [˚C]





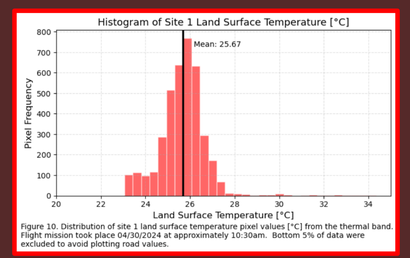


Figure 10. Distribution of site 1 land surface temperature pixel values [°C] from the thermal band. Flight mission took place 04/30/2024 at approximately 10:30am. Bottom 5% of data were excluded to avoid plotting road values.

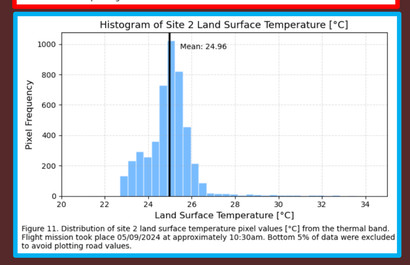
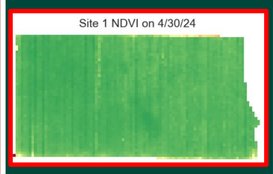


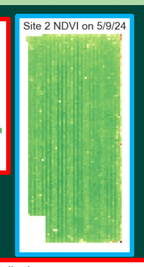
Figure 11. Distribution of site 2 land surface temperature pixel values [°C] from the thermal band. Flight mission took place 05/09/2024 at approximately 10:30am. Bottom 5% of data were excluded to avoid plotting road values.

Table 3. Site 1&2 Thermal Band Summary Statistics		
Land Surface Temp [˚C]	SITE 1	SITE 2
Minimum	23.06	22.70
Maximum	34.34	33.55
Mean	25.67	24.96
Standard Deviation	1.05	1.12

Figures 10-11 and Table 3 correspond with low LST where LE is high (leaf cooling mechanism) and high LST where LE is low (leaf water stress). Site 1 (top) has, on average, higher surface temperatures, aligning with high latent heat flux in site 1 (Figure 8).

Normalized Difference Vegetation Index (NDVI)





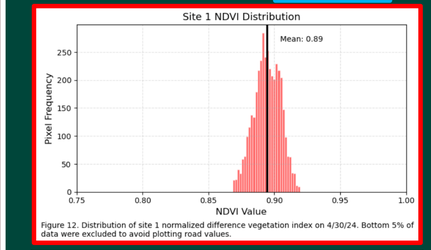


Figure 12. Distribution of site 1 normalized difference vegetation index on 4/30/24. Bottom 5% of data were excluded to avoid plotting road values.

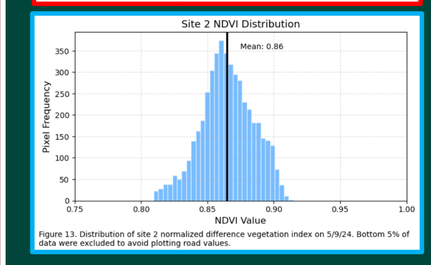


Figure 13. Distribution of site 2 normalized difference vegetation index on 5/9/24. Bottom 5% of data were excluded to avoid plotting road values.

Table 4. Site 1&2 NDVI Summary Statistics		
NDVI	SITE 1	SITE 2
Minimum	0.30	0.41
Maximum	0.92	0.91
Mean	0.89	0.86
Standard Deviation	0.026	0.034

Figures 12-13 and Table 4 summarize the NDVI distributions for each site. Site 2 has a wider distribution than site 1. This could be influenced by the almond phenological stage or almond tree species variability within site 2 vs. site 1.

Discussion

	SITE 1	SITE 2
Irrigation Distribution Uniformity (DU)	Lower DU; technicians found clogged emitters	Higher DU, more uniform irrigation
Latent Heat Flux (Evapotranspiration)	More heterogeneity	Smoother ET pattern
Land Surface Temperature	On average, slightly higher temperatures	Slightly lower temperatures with narrower distribution
Normalized Difference Vegetation Index	Narrower distribution of NDVI values	Wider distribution and some lower NDVI values
Conclusion	Lower distribution uniformity assessment results connect with heterogeneity in evapotranspiration and canopy temperature across a canopy.	

Future Work

- Complete more UAV missions paired with DU tests across the San Joaquin Valley (summer 2025) to capture a range of data points from poor to great DU
- Take steps toward automating the DU assessment to reduce the time cost
- Extend easy-to-use system to conservation districts, trade associations, and growers to increase DU assessment frequently and save water

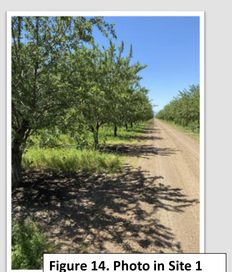


Figure 14. Photo in Site 1 almond orchard [6]

Acknowledgements

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